

TAK Server : Enabling 'X-Forwarded-For' Logging Procedure

TAK (Team Awareness Kit) Server's default HTTP access log records only the address of the upstream device that connects to it. When the server sits behind a load balancer or reverse web proxy, such as a BIG-IP performing SNAT, every HTTP log entry shows only the proxy's address, even though the proxy might be sending the X-Forwarded-For header. The log format is fixed in compiled Java code and cannot be changed through TAK CoreConfig.xml or TAK startup script options. This process modifies the one class that defines the Tomcat log format so that every entry also records any X-Forwarded-For value, allowing every request to be traced to its originating client. This process is useful for organizations that download pre-built RPMs from tak.gov and do not maintain their own TAK build server. This process has been validated against TAK Server [5.5-RELEASE-82](#) and should work for subsequent releases, provided that the TAK developers do not change the Tomcat architecture.

Tak build server or a non-Tak RHEL server tools preparation

1. Install the required packages

```
dnf install -y java-21-openjdk-devel unzip binutils
```

Windows client tools preparation

2. Download Recaf, a Java bytecode analyzer and disassembler: <https://github.com/Col-E/Recaf>
3. Download a Java Development Kit (required for Recaf): <https://www.oracle.com/java/technologies/downloads/>

Back up the current unmodified war

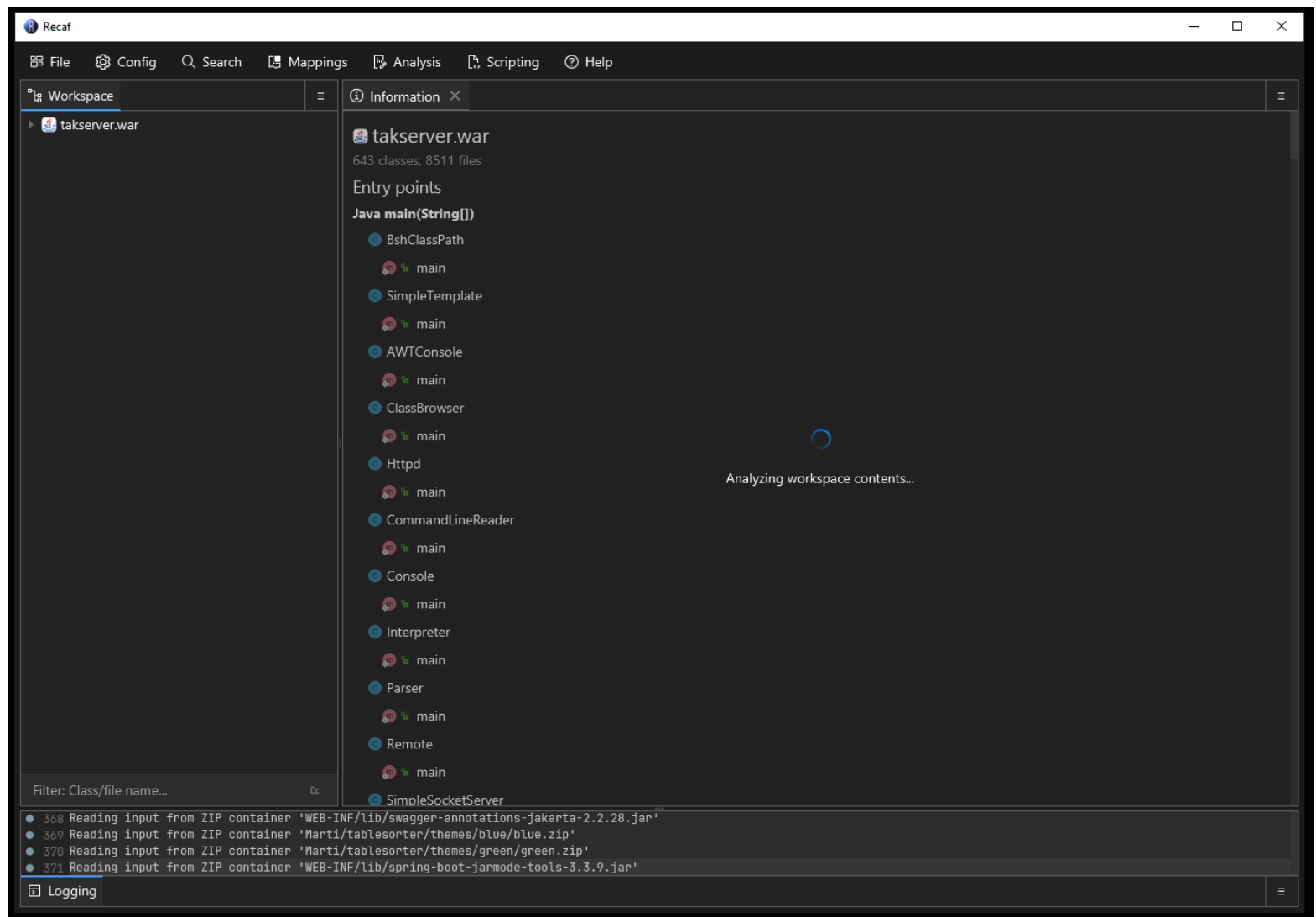
4. Copy the default war to a backup version

```
cp -a /opt/tak/takserver.war /opt/tak/takserver.war.backup
```

Process and modify the takserver.war and export a single modified class file

5. Copy takserver.war to the Windows client machine via scp
6. Open Recaf by double-clicking the Recaf jar file

7. Drag and drop the unmodified takserver.war into the Recaf window and wait until war analysis is complete



8. In the left workspace window, navigate to **Classes > tak > server > CustomizeEmbeddedTomcatContainer**

9. CustomizeEmbeddedTomcatContainer is the class that instructs Tomcat on how to log HTTP connection information. In the right panel find the line:

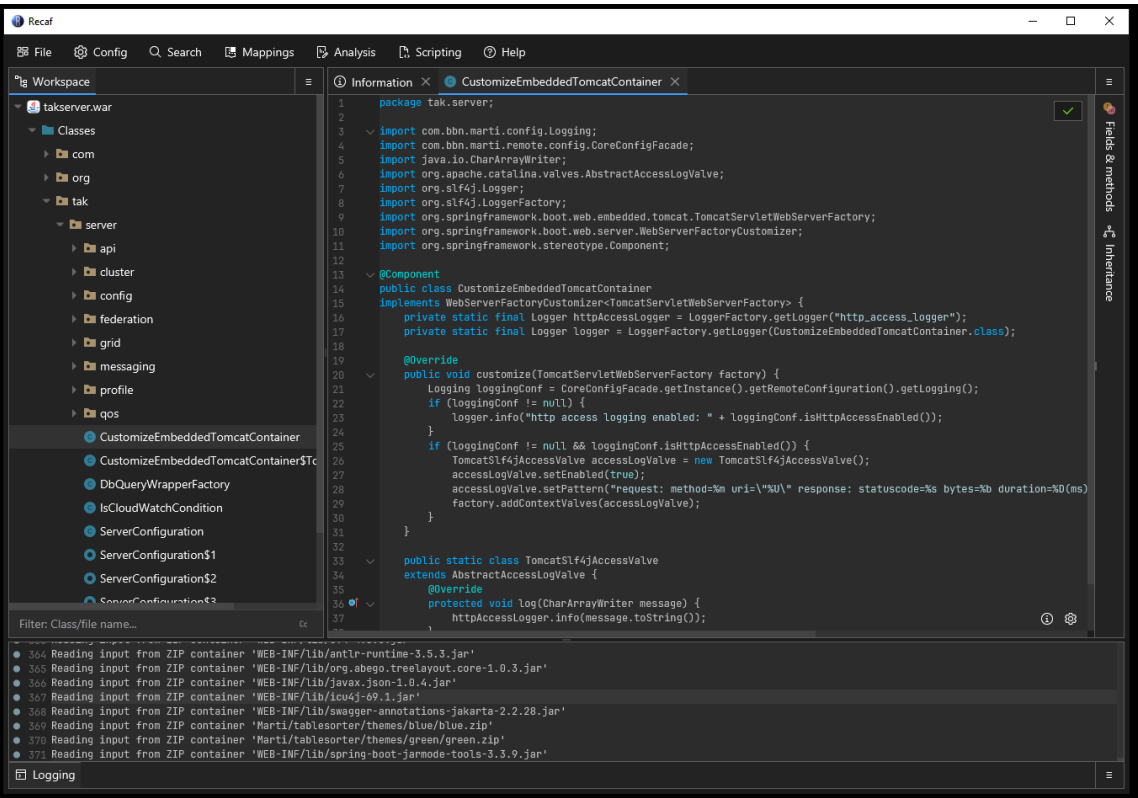
```
accessLogValve.setPattern("request: method=%m uri=\"%U\" response: statuscode=%s bytes=%b duration=%D(ms) client: remoteip=%a user=%{username}s useragent=\"%U-Agent{i}\"");
```

10. And modify it to include the incoming X-Forwarded-For header via: **xff=%{X-Forwarded-For}i**

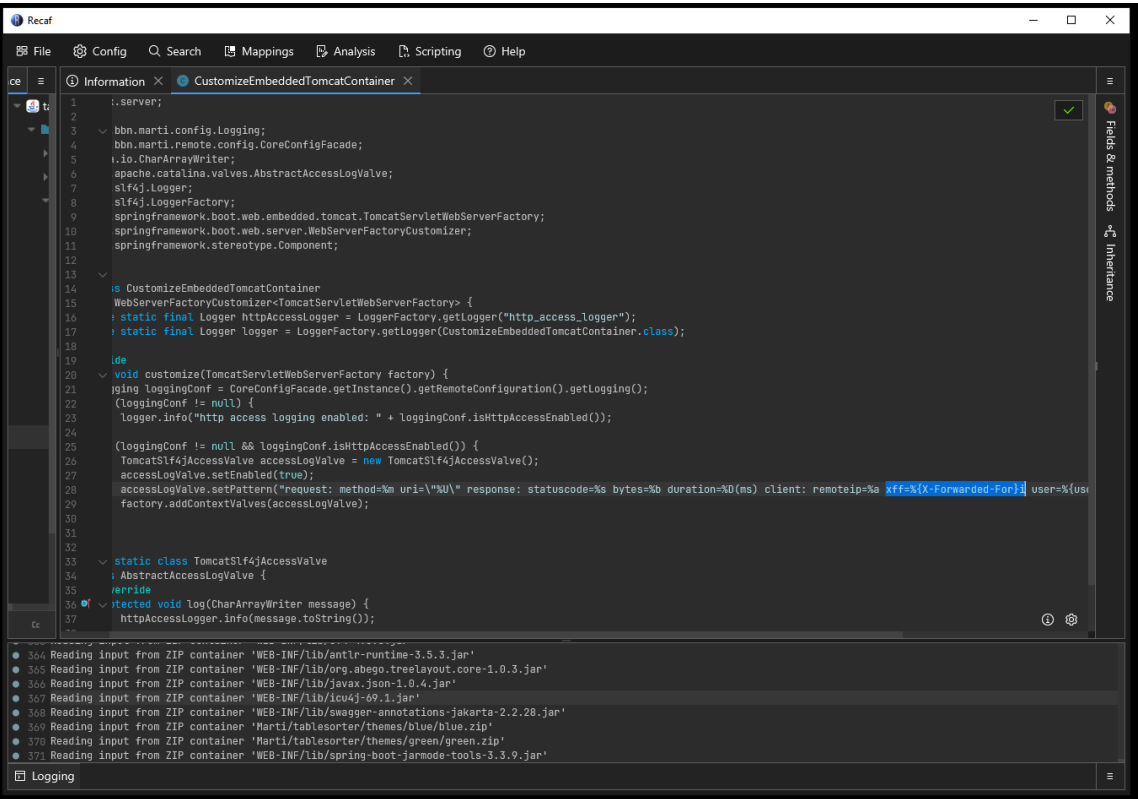
```
accessLogValve.setPattern("request: method=%m uri=\"%U\" response: statuscode=%s bytes=%b duration=%D(ms) client: remoteip=%a xff=%{X-Forwarded-For}i user=%{username}s useragent=\"%U-Agent{i}\"");
```

11. Do not make any other changes. Press Control-S to save the now modified class file.

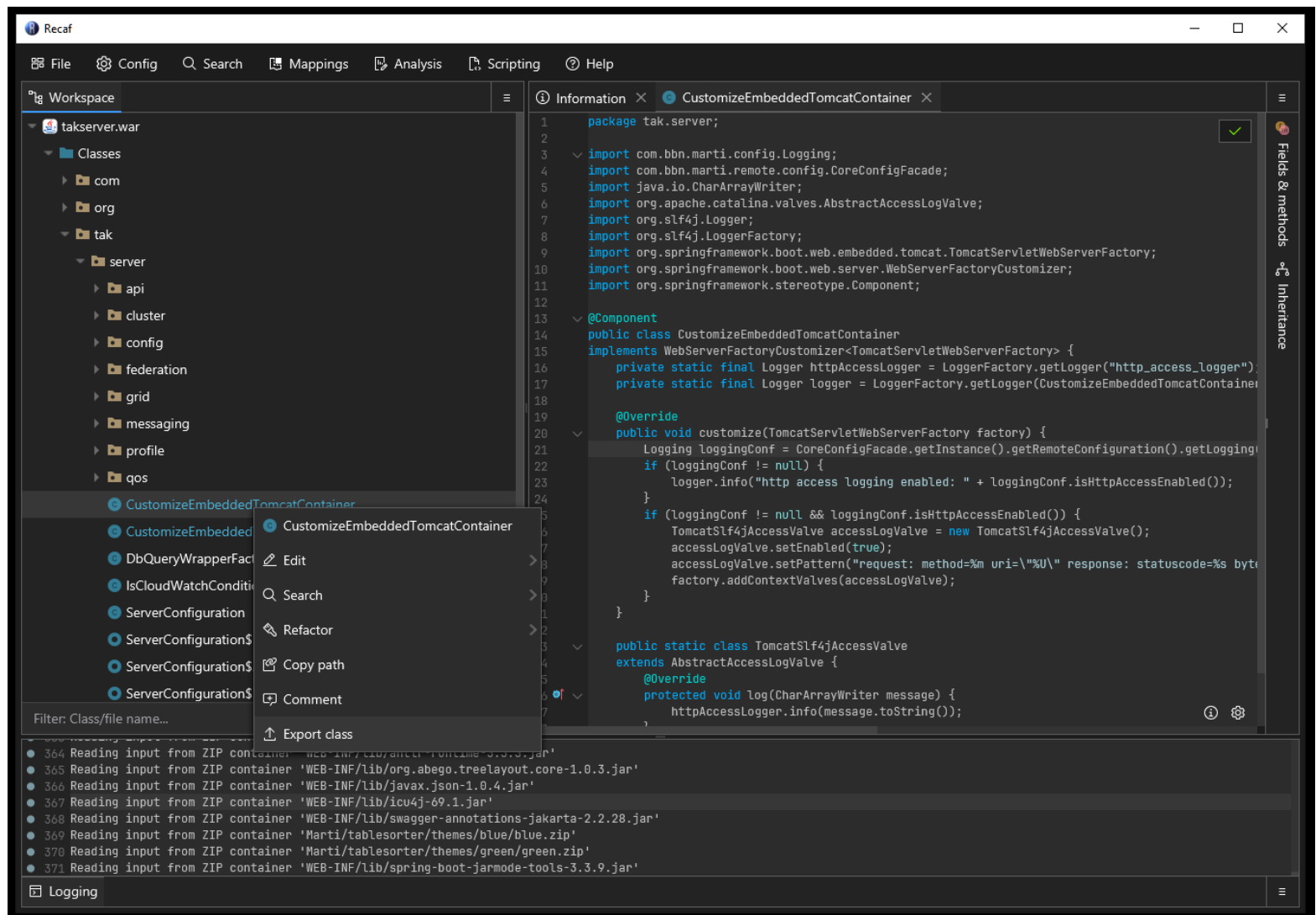
Original:



Modified:



12. Do not export the entire war using Recaf, this repackages the war in a layout the TAK Server cannot load. Right click on 'CustomizeEmbeddedTomcatContainer' and select **"Export class"**. Recaf will ask you where to save the 'CustomizeEmbeddedTomcatContainer.class' file. Ensure you do not edit this file in notepad or other text editor or you will cause bytecode corruption of the class file rendering it useless.



13. Copy takserver.war and the modified class file to the Tak build server or a non-Tak RHEL server, into the same directory

Patch the modified class file into the war

14. Run these commands in the directory holding takserver.war and the modified class file
- jar places a file in the war at the same relative path it has on disk, so the class **must** be moved into a temporary **WEB-INF/classes/tak/server** folder which builds the necessary internal pathing in the war during the patch

```
mkdir -p WEB-INF/classes/tak/server
mv CustomizeEmbeddedTomcatContainer.class WEB-INF/classes/tak/server/
jar uf takserver.war WEB-INF/classes/tak/server/CustomizeEmbeddedTomcatContainer.class
rm -rf WEB-INF
```

15. Verify the patch – The output **must be 1** before proceeding further

```
unzip -p takserver.war WEB-INF/classes/tak/server/CustomizeEmbeddedTomcatContainer.class | strings | grep -c 'xff='
```

Install the patched war

16. On the TAK server, stop the service *(ensure to wait until the server has completely shutdown)*

```
systemctl stop takserver
```

17. From the Tak build server or a non-Tak RHEL server, copy the patched takserver.war to the TAK server at /opt/tak/takserver.war

```
scp takserver.war root@<tak-server>:/opt/tak/takserver.war
```

18. On the TAK server, set ownership and start the service

```
chown tak:tak /opt/tak/takserver.war  
systemctl start takserver
```

Logging Verification

19. Review the access log

```
tail -f /opt/tak/logs/takserver-api-access.log
```

Expected entry:

```
... client: remoteip=<BIG-IP address> xff=<client address> user=... useragent="..."
```

Rollback

20. Restore the backup and restart TAK Server *(ensure to wait until the server has completely shutdown)*

```
systemctl stop takserver  
cp -a /opt/tak/takserver.war.backup /opt/tak/takserver.war  
systemctl start takserver
```